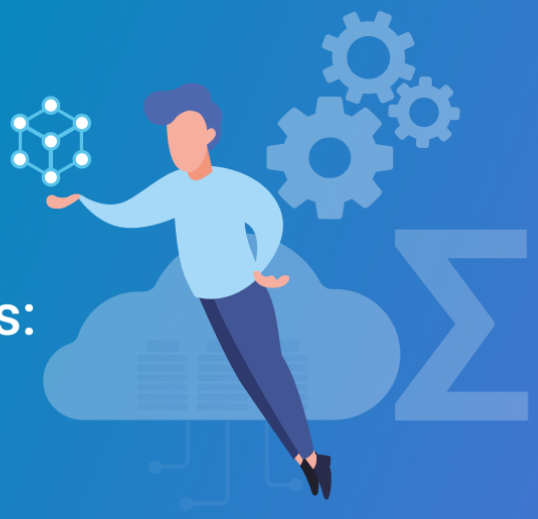
AKUR8



Rate Change and Dislocation Analysis:

Transparent ML as the path to accurate and actionable insights

The imperative of pricing updates

Newly emerging changes in risk and customer behaviour, a rapidly changing competitive environment together with major challenges due to the COVID-19 pandemic, all call for more competitive and accurate pricing. This increasing pressure for better pricing strategies ultimately must translate to changes in price away from the current strategy.

To build a new pricing strategy, an important piece of the equation is to have a reliable estimate of the risks within the current portfolio. New trends in risk, a shifting mix of characteristics in the portfolio, along with new sources of data, may uncover new buckets of high risk individuals for which a high premium increase should be enacted to maintain portfolio profitability and loss ratio.

Such disruptive price changes have a negative impact on customer retention. As companies have to manage their existing book of business, it is important to avoid, as much as possible, rate updates that generate a strong negative impact on retention, driving policies away with high premium increases. Furthermore, abnormal price changes may draw additional concerns from the regulators as they may receive complaints of unfair rate increases.

For these reasons, it is vital for actuaries to have complete control and understanding of the final rate changes that are applied to the current portfolio by a proposed pricing update.

The challenge of controlling rating disruptions

Control of rate changes is particularly challenging for complex insurance products. Several risks are covered by these contracts, and each risk requires its own specific modeling. When all these risks are combined, the cumulative effect may lead to disruptive rate changes; both disruptive increases that negatively impact retention, and decreases which negatively impact profitability.

To master the delicate balance between portfolio profitability and retention, it is key for insurers to directly review the impact of a planned rate change on their portfolio. When in the

process of updating their rates, insurers must analyse the impact of the projected change between old and new prices. This approach, commonly known as **"Dislocation Analysis"**, stands out as a must-have in order to directly and accurately understand the impact of price instability on renewing customers. Mitigation measures, such as manual rating adjustment to limit extreme price changes, should be taken based on direct price understanding, without over-reliance on a large number of models.

To date, **Dislocation Analysis** is often performed either in Excel or via an ad-hoc solution, created in-house. The pricing expert has to manually understand the portfolio segments which are excessively impacted by the new pricing strategy via a time-consuming variable by variable analysis. Then, several stakeholders, from the modeler themselves to the Product Owner have to define some rating adjustments, whose efficacy has to be validated by re-running the same lengthy and iterative analysis.

Empowering analysis with Transparent Machine Learning

The democratization of quick and efficient modeling brought about by Transparent Machine Learning can be of tremendous help in empowering Dislocation Analysis.

Transparent Machine Learning can provide direct and visual answers to key question surrounding Dislocation Analysis such as:

- What are the key characteristics describing the segments of the portfolio which are severely impacted by the price change?
- Are the price increases imposed on clients always well motivated?
- What are the best set of adjustments that could be made to minimise the negative impact of large price changes?

The dynamics of an integrated Dislocation Analysis, powered by Transparent Machine Learning, are straightforward and can be summarized as Select, Analyse and Action.

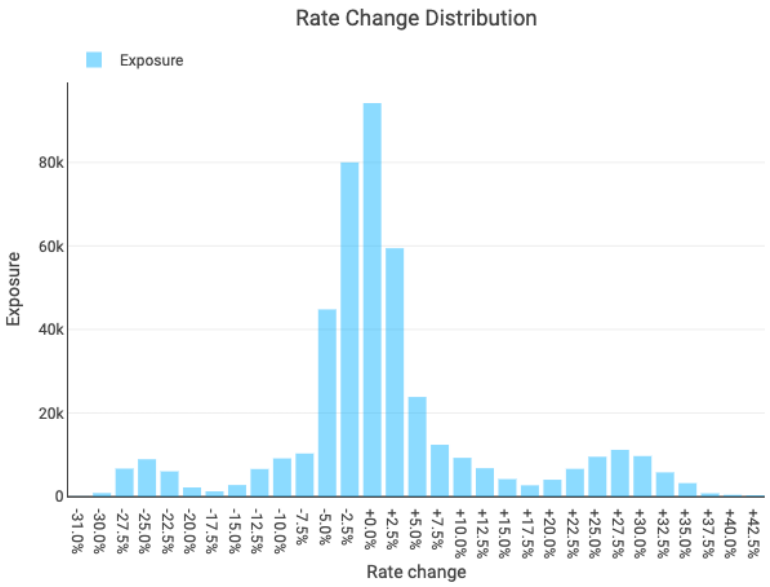
1. **Select:** The Dislocation Analysis starts with the analysis of the distribution of the rate changes or the loss ratios. More than often, these graphs highlight portions of the portfolio which can be problematic from a retention or profitability point of view. An integrated Dislocation Analysis workflow allows the pricing expert to easily select all the subset of clients whose proposed price change is potentially disruptive.
2. **Analyse:** Once a subset of the portfolio is highlighted, transparent Machine Learning allows the pricing expert to quickly find the root causes of the problems. The pricing expert can analyse the problem at various depths of sophistication; from a quick univariate exploratory analysis up to a more in-depth GAM modeling, isolating the individual effect of each of the different variables.
3. **Action:** Once the potentially disruptive portfolio segment is well defined through their rating variables, the pricing expert can define the right rating adjustments to mitigate the impact on retention or profitability. This set of modifications is provided by the outputs of the Transparent Machine Learning.

Once the 3 steps are completed, the analysis can be run again to refine the rating plan until the optimal tradeoff between retention and profitability is achieved. The level of sophistication of a Dislocation Analysis powered by Transparent Machine Learning allows for a faster and much more powerful detection of highly increased/decreased segments. This efficiency creates a much more precise price-update process, allowing higher price agility while avoiding all unnecessary disruption in the pricing strategy.

We provide two use-cases to showcase the Select, Analyse and Action flow applied on Dislocation Analysis.

Use Case: Rate Change

First, consider the use case of evaluating the individual rate change from the Current Premium to the new Proposed Premium. The Select step of the analysis starts by visualizing the distribution of the rate change (Premium vs Current) on our in-force portfolio.

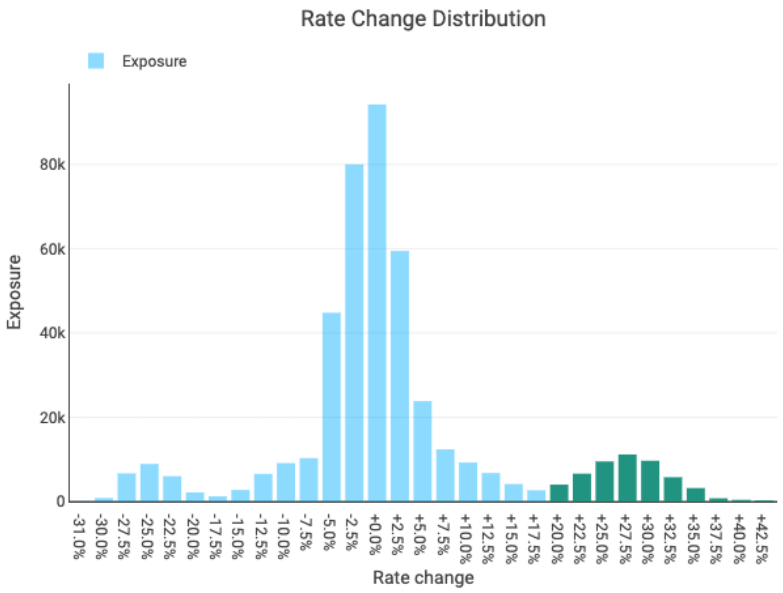


In the Rate Change Distribution figure above, the x-axis represents the rate change impact and the y-axis represents the number of clients within each rate change bucket. The left side of the graph represents the customers who will expect a price decrease, while the right represents those whose tariff will be increased.

Source : Automated output of Akur8's Dislocation Analysis.

Under this rate change, there is a significant portion of the portfolio whose price will be increased by a very high amount.

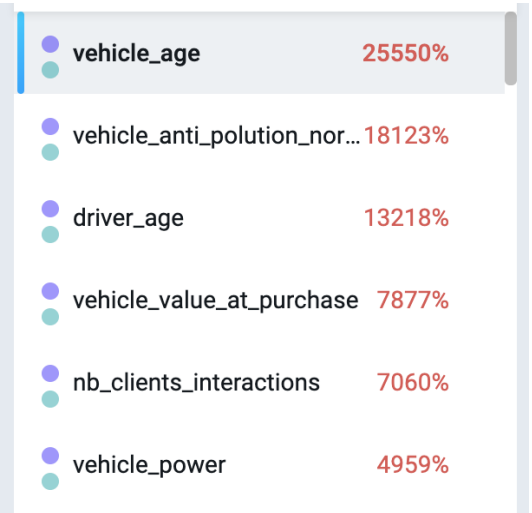
It could be decided that the goal is to identify the profiles whose price will be increased by more than 20%. The pricing expert can highlight that segment directly on the rate change graph to start the investigation.



Rate Change Distribution graph highlighting the part of the portfolio whose price will be increased by more than 20%.

Source : Automated output of Akur8's Dislocation Analysis.

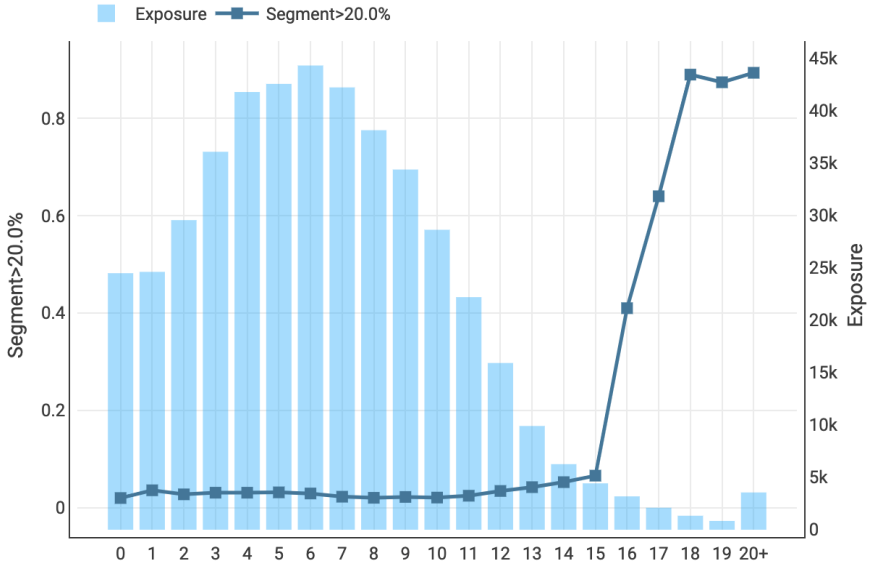
Quickly the pricing expert can obtain an automatically guided univariate analysis.
First, the rating plan variables are ranked by their discriminative power, allowing the pricing expert to quickly spot the set of variables which are driving the price increase, among the high number of factors that build the rating plan.



The above figure displays the top 6 most significant variables that describe the segment of people with a high price increase, as selected above.

Source : Automated output of Akur8's Dislocation Analysis.

Then, those driving factors can be visualised to quickly describe the key features severely impacted by the new premiums.



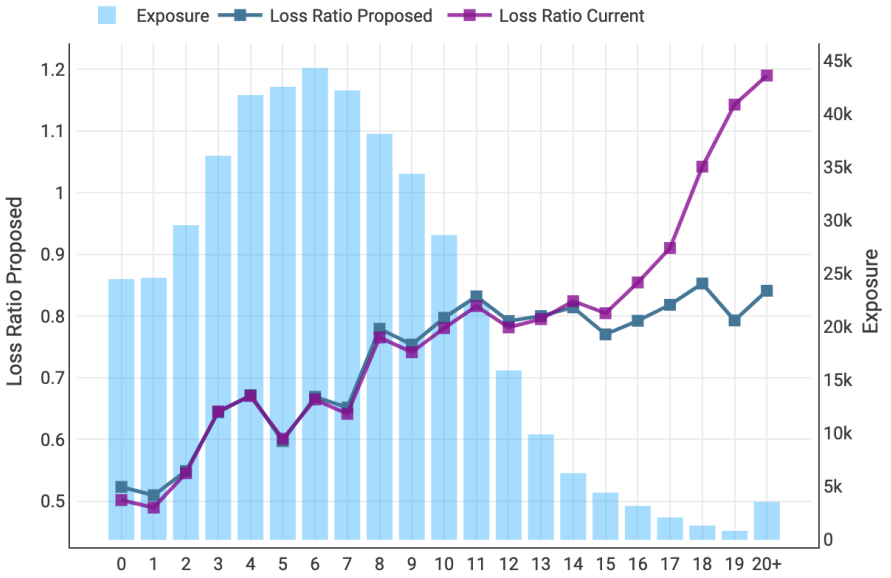
The figure above shows the most important variable to describe the disrupted segment, the vehicle_age. The x-axis represents the age of the vehicle in years, and the y-axis represents the share of clients within each group whose price will be increased by more than 20%.

It is clear that the new rate change severely affects the part of the portfolio driving older cars: 90% of people driving an old vehicle will have their rates augmented by more than 20%.

Source : Automated output of Akur8's Dislocation Analysis.

Furthermore, additional KPIs can be displayed to check whether the rate changes are well-motivated, taking profitability into account.

High price increases lead to a strong attrition of existing customers. This phenomenon is acceptable, and less important, when applied to unprofitable segments (as the unprofitable customers are usually less price-sensitive, and it is more acceptable to trigger higher lapse on these segments).



The figure above displays the loss ratios under the old current (purple) and the new proposed (blue) premium on the Vehicle Age. Older vehicles were indeed unprofitable under the current premiums, having observed loss ratios above 100%. The changes applied by the new proposed premium effectively act on this level of unprofitability.

While the price update may be too drastic and could be more moderate, it seems the update is indeed well-motivated.

Source : Automated output of Akur8's Dislocation Analysis.

This univariate analysis offers a first degree of sophistication to understand the impact of the rate change. If required, a more comprehensive description of the root causes can be achieved through more complex modeling of the identified segments of interest. Without an integrated Dislocation Analysis tool, a modeling effort to effectively detect the root causes of the rate disruptions is often overlooked due to the time and resource intensive nature of building GAMs and GLMs from scratch. Integration and Transparent Machine Learning breaks down such barriers. A fast and automated model building process opens the door for quick and powerful investigations on top of the univariate analysis, providing two main benefits.

Firstly, modeling allows for a more exact detection of the most important factors which are driving the rate change, by isolating the individual effect of each variable.
Secondly, a model will display exact rating adjustment that would reduce the effects of rating disruption, providing a strong guidance for the adjustment to take for further iterations.

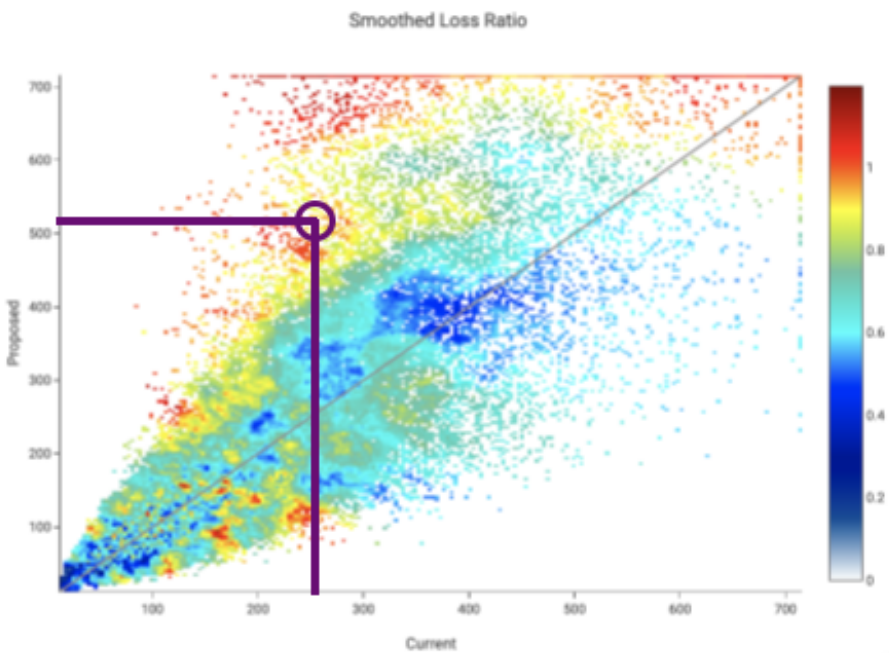
After the **Analyse** step of the Dislocation Analysis, the variables and the segment to be adjusted in the rating plan are defined with surgical precision. The pricing expert can now apply those adjustments in each coverage of the rating plan, leading to a new proposed premium which will effectively control the rate disruption previously observed.

Use case: Portfolio profitability / Loss Ratio

In the previous use-case, Rate Change, the **Select, Analyse** and **Action** flow showed how quickly the pricing expert can move from detecting a problematic segment in the portfolio, to defining the actions needed to mitigate renewal disruptions.

The same kind of flow can be applied simultaneously to rate change and loss ratio analysis, to highlight and detect non-profitable segments that can arise due to a new rate change.

A first step of loss ratio analysis is illustrated in the graph below. It shows the smoothed loss ratio of customers targeted by a price change from the current price to the new projected price.



The current price is displayed along the x-axis. The new projected price is displayed along the y-axis. Color indicates the loss-ratio (red dots represent unprofitable high loss-ratio segments while blue dots represent profitable low loss-ratio segments).

Here, some customers highlighted are currently priced at $250 (x-axis) and will be priced at $500 (y-axis) in the analysed strategy; this price change is motivated by the very high loss ratio predicted for these clients (as they appear in red).

Source : Automated output of Akur8's Dislocation Analysis.

The illustration here provides a holistic view of the impact of price changes on policyholders' profitability.

Customers below the diagonal receive a price decrease while those above suffer a price increase (those far apart from the diagonal receive a very strong price change while those on the diagonal are renewed with a price similar to their current one).

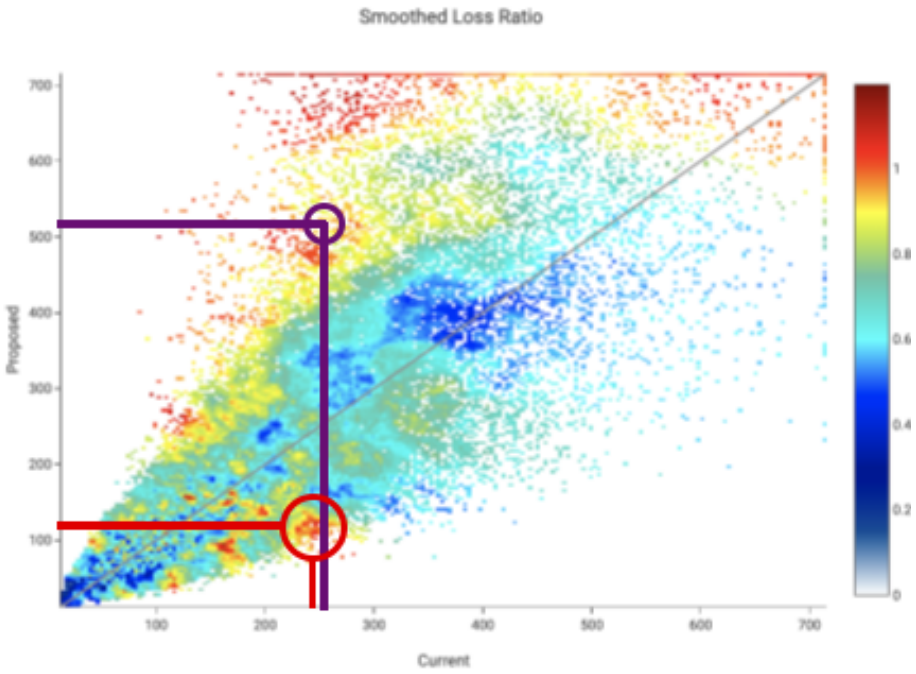
These price changes can be motivated by profitability considerations; low profitability policyholders (displaying high loss ratios, in red), would be expected to receive significant

price increase - and appear above the diagonal - while high profitability ones could receive a price decrease.

This graph allows the pricing expert to directly spot the segments most impacted by price changes, and, in particular, detect whether these price changes are motivated by profitability improvements.

This global review of price changes should be deepened by further analysis:

- The insurer can spot anomalies in the price changes, for instance segments with price decreases on unprofitable clients, as illustrated in the chart below (light red)
- The insurer can further identify which segments are describing these clients and should be re-priced before deploying the proposed strategy



The points identified above display a strong price decrease (from $230 to $110), while having a high loss-ratio (above 100%, as identified by their red color). Applying a strong price decrease on unprofitable clients might not be an efficient strategy, and deeper investigation on this segment to understand the motivation of this price update is necessary.

Source : Automated output of Akur8's Dislocation Analysis.

The in-depth analysis can be achieved following a process similar to the one described above, by:

- Selecting the suspicious segment
- Identifying, thanks to Transparent Machine Learning technologies, which variables best describe this segment
- Visualizing the relationship between these variables of interest and the segment or the loss-ratio

- If needed, taking actions to offer a different price update to the customers identified by the analysis.
- Overall, leveraging Dislocation Analysis allows insurers to directly visualize the expected loss ratio impact of their projected price changes. Furthermore, the **Select, Analyse** and **Action** flow allows seamless detection of the segment of the portfolio which are mis-priced, providing a more accurate picture of what the “right” pricing approach for their customers should be.

Conclusion

Renewing the pricing strategy to adjust to market and risk changes is essential for insurers and requires them to find the optimal tradeoff between rate disruptions and profitability. These matters are the key objectives of the Dislocation Analysis. Empowered by Transparent Machine Learning, the pricing expert can quickly select and analyse the portfolio segments which are negatively impacting the insurer’s business and leverage the insights to build the best possible premiums.

For more information, visit www.akur8.com or contact us at contact@akur8.com.